The empirical recurrence for OEIS A214103, proved by transfer matrix

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2 September 2026

Abstract

OEIS A214103 counts the proper colourings of a grid $4 \times (n+1)$, wrapped in the growing direction, with at most 3 colours and counted up to renaming the colours, and carries an empirical recurrence of order 11 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. Counting up to renaming is a fixed rational combination of ordinary colouring counts; each of those is a walk count; and because permuting the colours preserves everything in sight, the walk may be run on the colour-patterns rather than on the colourings, which is what brings the state count down to $S = 98$ and makes the exact residual test affordable.

2020 Mathematics Subject Classification. 05A15, 05C15, 05C30.

1 The conjecture

OEIS A214103 is "Number of 0..2 colorings of a 4X(n+1) array circular in the n+1 direction with new values 0..2 introduced in row major order.". It has offset 1 and begins

$$27, 8, 771, 1080, 29540, 79968, 1241355, 4807928, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 8*a(n-1) +26*a(n-2) -304*a(n-3) +304*a(n-4) +2016*a(n-5) -3991*a(n-6) -2616*a(n-7) +9702*a(n-8) -2064*a(n-9) -6120*a(n-10) +2880*a(n-11)

As of the "Last modified" line on the live entry (Oct 03 2025 16:51:20, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

A *colouring* here is proper: cells adjacent horizontally or vertically carry different values, and "circular in one direction" means the two ends of that direction are adjacent as well. The check that this is the right reading is the entry's own first term for the one-row case: a cycle of W cells has $(i-1)^W + (-1)^W(i-1)$ proper colourings over i colours, which at $W = 6$, $i = 3$ is 66, and $66/3! = 11$ is what the corresponding entry of this family publishes.

The clause "new values 0..2 introduced in row major order" says the array is the canonical representative of its equality pattern, so what is counted is patterns, not arrays. Write N_j for the number of admissible patterns using exactly j colours and L_i for the number of admissible colourings over an alphabet of i colours. A colouring over i colours is a pattern together with an injection of its colours into the alphabet, so

$$L_i \;=\; \sum_j N_j \, i(i-1)\cdots(i-j+1),$$

a triangular system with unit diagonal, hence invertible over $\mathbf{Q}$. The entry counts $a = N_1 + \cdots + N_3$, so

$$a \;=\; \sum_{i=1}^{3} w_i\, L_i$$

for one fixed rational vector $w = \mathbf{1}^\top F^{-1}$, F the matrix of falling factorials. Being properly coloured is a statement about equalities alone, so it is invariant under permuting the colours and the reduction is legitimate.

3 Each L_i is a walk count

A column of the array is a proper colouring of the path P_4 and two consecutive columns must differ in every one of the 4 positions; the wrapping makes the last column adjacent to the first. A colouring of the array is therefore a CLOSED walk of $m = n+1$ steps in the digraph T_i whose vertices are the proper colourings of P_4 over i colours and whose edges join colourings differing everywhere, so

$$L_i(n) \;=\; \mathrm{tr}\; T_i^{\,m}, \qquad m = n+1.$$

A trace is not $\iota^\top M^m \tau$ for fixed vectors, and computing it as a sum over all starting vertices is far too expensive. Lemma 1 replaces it by a handful of walks with fixed ends.

Lemma 1. *Let G be a group acting on the vertex set of a digraph and preserving its edges, and let M be the adjacency matrix. Then $(M^m)_{ss} = (M^m)_{\sigma s,\sigma s}$ for every $\sigma \in G$, and hence*

$$\mathrm{tr}\, M^m \;=\; \sum_{\textit{orbits } O} |O|\; (M^m)_{s_O s_O},$$

one representative s_O per orbit. For T_i with $G = S_i$ permuting the colours, the orbits are the equality patterns and the orbit of a colouring using j colours has $i(i-1)\cdots(i-j+1)$ members.

Proof. σ permutes the vertices and carries walks to walks bijectively, so the closed walks of length m at s correspond to those at σs. Summing the diagonal over each orbit gives the displayed identity. Permuting colours preserves "differ in every position", so S_i acts on the colourings of P_4 preserving T_i; two colourings lie in one orbit exactly when they have the same equality pattern, and a pattern using j colours is realised in $i(i-1)\cdots(i-j+1)$ ways. ☐

Lemma 2. *Let a group G act on the vertices of a digraph preserving its edges, and let v be constant on the G-orbits. Then Mv is constant on the G-orbits, and for a representative s of the orbit O,*

$$(Mv)(O) \;=\; \sum_{O'} A_{OO'}\, v(O'), \qquad A_{OO'} = \#\{t \in O' :\; s \to t\}.$$

So an iteration started at a G-invariant vector may be carried out on the orbits.

Proof. $\sigma \in G$ is an edge-preserving bijection, so it carries the out-neighbours of s onto those of σs; with v constant on orbits, $(Mv)(s)$ and $(Mv)(\sigma s)$ are the same sum. The count $A_{OO'}$ is likewise independent of which representative of O is used. ☐

Both endpoint vectors above are invariant under the relevant group — the all-ones vectors under all of S_i, and, for a diagonal entry $(T_i^m)_{ss}$, the vector e_s under the subgroup fixing s — so the whole computation runs on orbits. That is the entire reason the state count is $S = 98$ rather than the number of colourings.

Assembling the pieces, $a(n)$ is $\iota^\top M^j \tau$ on the disjoint union of the orbit digraphs for $i = 1, \ldots, 3$, with ι carrying the weights w_i cleared of denominators; the walk index j and the entry's n differ by a constant, read off by matching the entry's published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 3. *Let $q(t) = t^{11} - \sum_i c_i t^{11-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+11} - \sum_i c_i M^{j+11-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 4.

$$a(n) \;=\; 8\,a(n-1)+26\,a(n-2)-304\,a(n-3)+304\,a(n-4)+2016\,a(n-5)-3991\,a(n-6)-2616\,a(n-7)+9702\,a(n-8$$

for every $n > 10$.

Proof. The vector $w = q(M)\tau$ was formed by 11 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 10 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. The rational weights w_i are cleared by a common denominator before any of this, so the whole computation is over the integers. □

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 20 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 3 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A214103.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[3] N. Biggs, *Algebraic Graph Theory*, 2nd ed., Cambridge University Press, 1993. (Chromatic polynomials and transfer matrices.)

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.